

Introduction to Health Economics

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- Why buy insurance?
 - Demand for insurance driven by the fear of the unknown
 - Hedge against risk - the possibility of bad outcomes
 - Purchasing insurance means forfeiting income in good times to get money in bad times
 - If bad times avoided, then money lost
 - Ex: The individual who buys health insurance but never visits the hospital might have been better off spending that income elsewhere.
 - It is risk aversion that drives demand for insurance.

- Introduce the simplest possible model that illustrates the notion of risk aversion and the demand for insurance.
- Consider an individual who cares about the income she earns and nothing else i.e. $U(I)$ with the following properties:
 - Utility increases with income: $U'(I) > 0$
 - Marginal utility for income is declining: $U''(I) < 0$

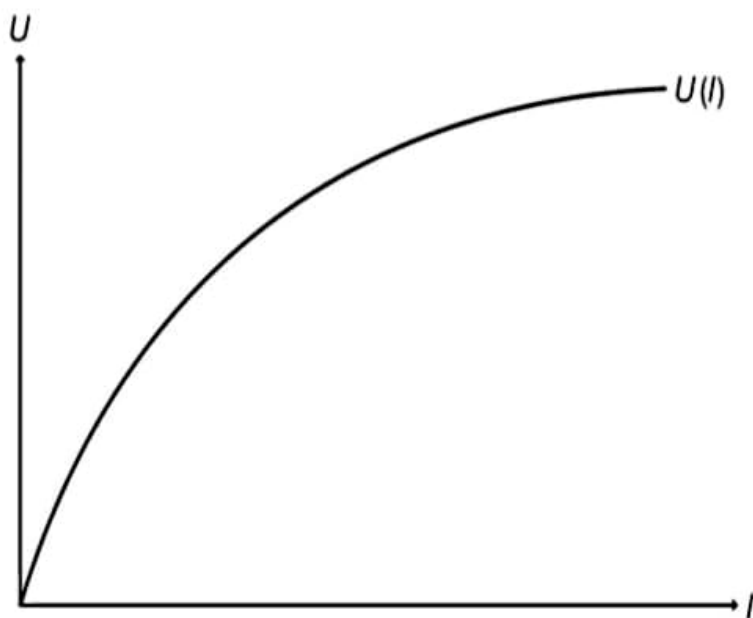


Figure 1: An income–utility diagram.

- Adding uncertainty to the model:
 - An individual does not know whether she will become sick, but she knows the probability of sickness is p between 0 and 1
- If she gets sick, medical bills and missed work will reduce her income
 - I_S - income if she does get sick
 - I_H - income if she remains healthy where $I_H > I_S$
- Expected value of the income

$$E[I]_p = p * I_S + (1 - p) * I_H \quad (1)$$

- Assumption: Individual is risk reverse. Risk aversion:
 - Concave utility function - $U'(I) > 0$ & $U''(I) < 0$
 - Prefer certain outcomes to uncertain ones with the same expected income.
 - Example - Out of two possible options:
 - A: a lottery that awards INR 500 with probability 0.5 and INR 0 with probability 0.5
 - B: a check for INR 250 with probability 1
 - Risk averse individual will choose B over A.
 - Prefers the utility from expected income to the expected utility from uncertain income

$$U(E[I]) > E(U[I]) \quad (2)$$

$$E(U(I)) = p * U(I_S) + (1 - p) * U(I_H) \quad (3)$$

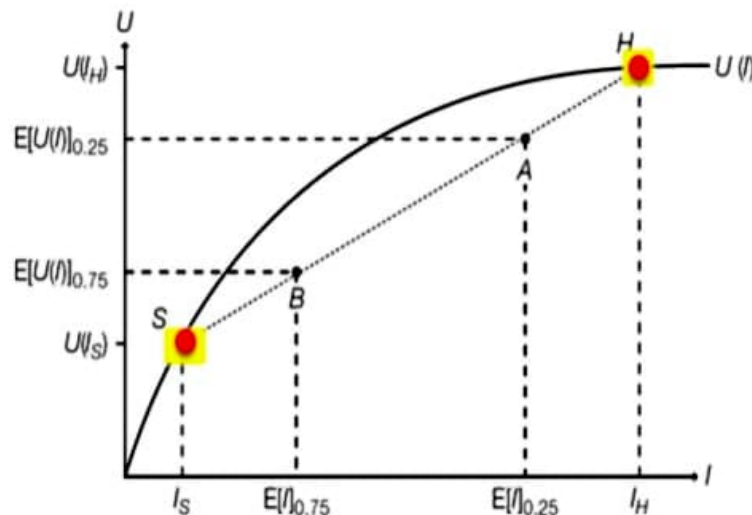


Figure 2: Utility Function

- Consider a case where the person is sick with certainty ($p = 1$):
 - $E[U] = U(I_S)$ equals the utility from certain income I_S (Point S)
- Consider the case where person has no chance of becoming sick ($p = 0$):
 - $E[U] = U(I_H)$ equals the utility from certain income I_H (Point H)

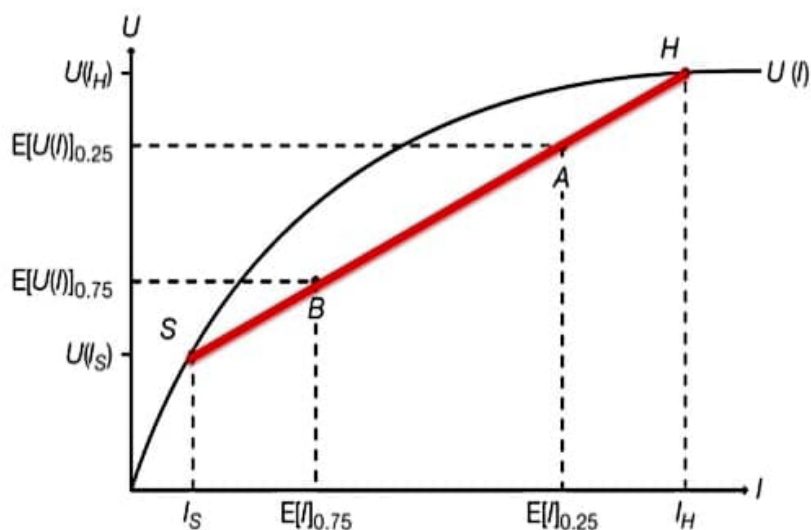


Figure 3: Utility Function

- For p between 0 and 1, expected utility falls on a line segment between S and H

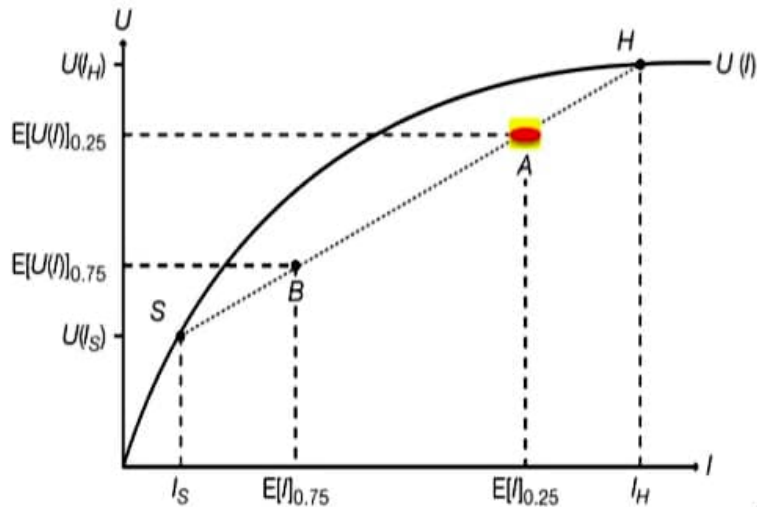


Figure 4: Utility Function

- For $p = 0.25$, person's expected utility is at point A
- For $p = 0.75$, person's expected utility is at point B

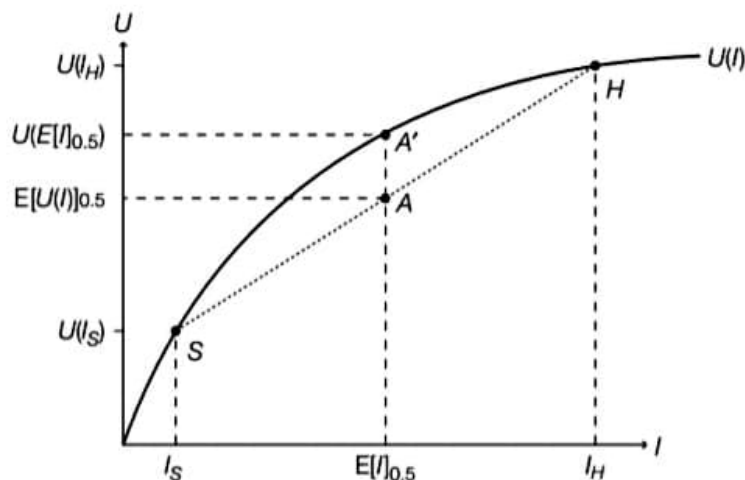


Figure 5: Utility Function

- For $p = 0.5$, person's expected utility is at point A
- whereas utility from expected income at $p = 0.5$ is at point A'. This value is the utility that would result if the individual could earn $E[I]_{0.5}$ with certainty.
- In our model, a person is either fully well or fully sick but never halfway between.
- This means that individuals cannot achieve the higher utility at A' on their own, even though risk averse people would prefer it.

- A risk-averse individual seeking to achieve A' would need to somehow send money from one potential self in the world where she stays healthy to her other potential self in the world where she becomes sick.
 - One can do it through a time machine may be???
 - or an insurance contract can help.
- A basic health insurance contract from an insurance company
 - Customer pays an upfront fee ' r ' - insurance premium.
 - If the customer falls sick then the customer receives q - the insurance payout
 - If healthy, customer receives nothing
 - Customer's final income with insurance:
 - Sick: $I'_S = I_S + q - r$
 - Healthy: $I'_H = I_H + 0 - r$
- Goal of a risk averse individual: Avoid uncertainty and get the expected income $E[I]_p$ with certainty no matter she is sick or healthy. As, A' gives her higher utility relative to A .

$$I'_H = I'_S = E[I]_p \quad (4)$$

- Design an insurance contract such that (4) holds true.

- Let $r = p \cdot q$ and $q = I_H - I_S$
- Then, in the healthy state: $I'_H = E[I]_p$
- In the sick state: $I'_S = E[I]_p$
- Verify.
- With an insurance contract: $r = p \cdot q$ and $q = I_H - I_S$, the individual can receive $E[I]_p$ with certainty.
- With the insurance contract, the individual's utility increases to A' even though her income does not.
- The insurance contract creates utility seemingly out of nowhere; simply by reducing uncertainty and making the risk-averse individual better off.
- The insurance contract acts as an instrument that transfers income from the healthy state of the world to the sick state.

- What can we interpret from $r = p^*q$ and $q = I_H - I_S$?
- Expected profit of the insurance company:

$$E[\Pi(p, q, r)] = (1 - p)r + p(r - q) \quad (5)$$

- If $E[\Pi(p, q, r)] = 0$ then $r = p^*q$. Expected profits comes down to 0.
- This is also called Actuarially fair insurance contract.
- An insurance contract which yields positive profits is called unfair insurance. $E[\Pi(p, q, r)] > 0 \Rightarrow r > p^*q$.
- An insurance contract will be actuarially fair in a perfectly competitive insurance market where profits will equal zero.
 - If profits were positive, new entrants would compete away those profits until all the firms left in the market would be making zero profits.
 - If the profits were negative, then the insurer is giving money away to customers in the long run and will go out of business. Firms leave the market until profits reach zero.

- Actuarially fair insurance, in a sense, is also free. The customer's expected income does not change from buying the contract, so she effectively pays nothing for it.
- However, insurance markets are not perfectly competitive and hence insurance companies generate positive expected profit by selling actuarially unfair contract
- The higher r rises above pq , the pricier the contract is and the more unfair it becomes.
- Unfair contracts reduces the expected income to be generated after buying insurance. Verify.
- Risk-averse consumers may still be willing to pay positive prices for unfair insurance contracts if doing so sufficiently reduces their uncertainty.

- Interpretation of second condition: $q = I_H - I_S$ implies that there is no income uncertainty after insurance.
- $I'_H = I'_S \Rightarrow q = I_H - I_S$
- This property is known as state independence, because the individual's income no longer depends on her health status.
- An insurance contract that achieves state independence completely eliminates income uncertainty and is called a full insurance contract.
- Under partial insurance, income in the sick state even with insurance is still lower than income in the healthy state ($I'_H > I'_S$), but some income is still transferred from the healthy to the sick.
- The size of the payout q determines the fullness of the insurance contract. A contract with a payout that fully covers the spread between I_H and I_S is full, while contracts with payouts that do not fully cover this difference are partial.

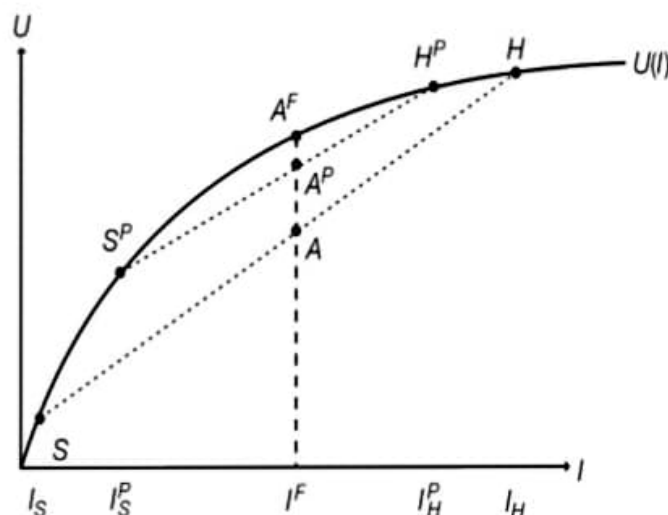


Figure 6: Comparing full-fair, partial-fair, and no insurance.

- For any risk-averse individual, actuarially fair and full insurance contract offers the most utility. Hence, it is called the ideal insurance contract.
- The fairness of a contract is considered as its effective price whereas fullness of a contract as its effective quantity.

- Comparing different insurance contracts:

	Fair	Unfair
Full	$r = pq$ $q = I_H - I_S$	$r > pq$ $q = I_H - I_S$
Partial	$r = pq$ $q < I_H - I_S$	$r > pq$ $q < I_H - I_S$

Figure 7: Comparing full-fair, partial-fair, and no insurance.

- At a fixed level of fairness (which we can think of as a fixed price), full insurance is preferable to partial insurance because it provides certainty about income.
- At a fixed level of fullness (or a fixed quantity), however, fair insurance is preferable because it costs less than unfair insurance and delivers the same benefit.

- Comparing non-ideal contracts:

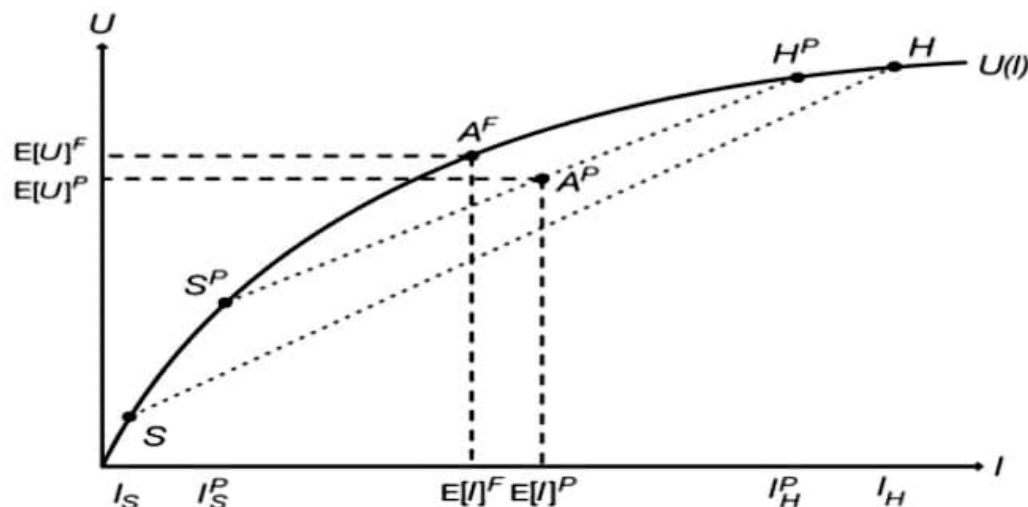


Figure 8: Two non-ideal contracts.

- A^F – Full but actuarially unfair contract
 - A^P – Partial but actuarially fair contract
- In this case, $U(A^F) > U(A^P)$
 - Even though A^F is actuarially unfair, its relative fullness (i.e. higher payout) makes it more desirable
 - However, if contract A^F became more unfair, then expected income $E[I]$ falls and A^F may generate less utility than A^P

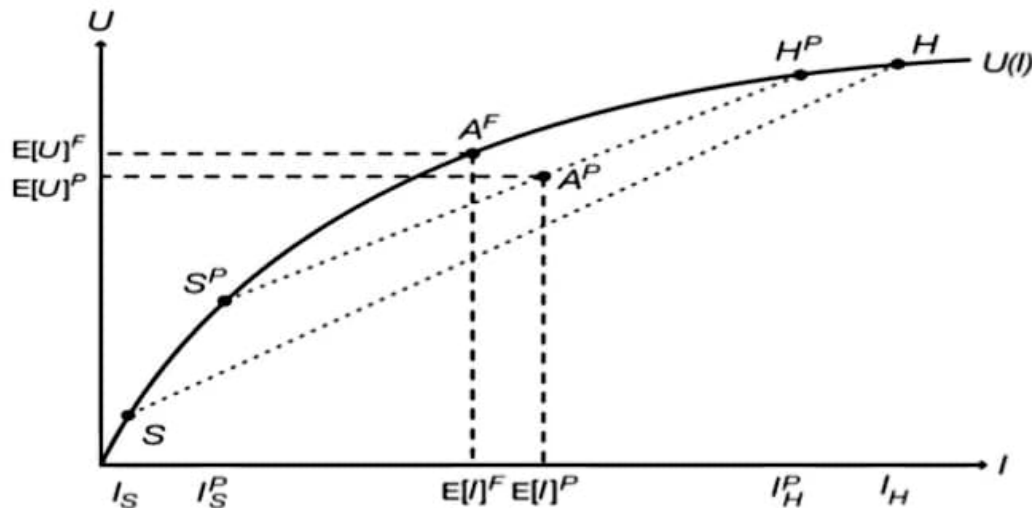


Figure 9: Two non-ideal contracts.

- Similarly, A^P may become more full by increasing its payout
 - At some point, this consumer will be indifferent between the two contracts i.e. new partial contract and A^P .
- Conclusion - Demand for insurance is driven by the tendency of risk aversion.
- Risk aversion can also explain
 - Large family sizes
 - Portfolio diversification.